order, with left justification, such that the first bit of the prefix is in the most significant bit of the first octet. Encoding SHOULD use the least number of bytes to encode the prefix but MAY include unused trailing zeroes.

Examples of encoding:

- Preferred minimal encoding: Prefix `2001:0DB8:0:CD30::/60` → 9 octets → `3C20010DB80000CD30`
- Preferred minimal encoding: Prefix `2001:0DB8:BB00::/40` → 6 octets → `2820010DB8BB`
- Allowed non-minimal encoding: Prefix `2001:0DB8:BB00::/40` → 7 octets → `2820010DB8BB00`

7.19.2.45. Hardware Address Type

The Hardware Address data type SHALL be either a 48-bit IEEE MAC Address or a 64-bit IEEE MAC Address (e.g. EUI-64). The order of bytes is Big-Endian or display mode, where the first byte in the string is the left most or highest order byte.

7.19.2.46. SemanticTagStruct Type

This data type SHALL be represented by the following structure:

ID	Name	Type	Constraint	Quality	Fallback	Access	Conformance
0	MfgCode	vendor-id		X	NULL		M
1	NamespaceID	namespace					M
2	Tag	tag					M
3	Label	string	max 64	X	NULL		MfgCode != NULL, O

MfgCode Field

If the MfgCode field is not NULL, it SHALL be the Vendor ID of the manufacturer who has defined a certain namespace and the NamespaceID field SHALL be the ID of a namespace defined by the manufacturer identified in the MfgCode field.

If a manufacturer specific Tag field is indicated in a list of SemanticTagStruct entries, the list SHALL include at least one standard tag which is not from any manufacturer's namespace. A standard tag is a tag from a common namespace, a derived cluster namespace, or an applicable device-specific namespace.

If MfgCode is NULL, the NamespaceID field SHALL indicate a standard namespace.

NamespaceID Field

The NamespaceID field SHALL identify a namespace.

The common and device-specific semantic tag namespaces are listed in StandardNamespaces.

